

Hybrid Inverter Super-Backup Function A special EPS mode for battery ready



Storage&Battery

Introduction

This article will introduce the Super-Backup Function on Solax Hybrid inverters. With the Super-Backup function, during black-out the inverter can supply the backup loads even if there are no batteries installed.

Different from the normal backup solution with batteries, the **Super-Backup** solution, during **EPS Mode**, hybrid inverters export power only from PV panels (without batteries). As the **PV input power** is not stable, there will be some limitation. In order to avoid the overload^①, the **EPS** ports (back-up loads) must be followed:

1. For single-phase inverters(X1-Hybrid-G4 as an example), the output power of **EPS** ports is restricted to half (50%) of the nominal output power of **EPS** ports (with battery). For more details in the Table1 below.

Table1. EPS output of X1-Hybrid G4 (without battery)

EPS output (without battery)	X1 Hybrid 3.0	X1 Hybrid 3.7	X1 Hybrid 4.6	X1 Hybrid 5.0	X1 Hybrid 6.0	X1 Hybrid 7.5
Nominal output power [W]	1500	1840	2300	2500	3000	3750
Peak apparent power [VA] (130% overload)	1950,10s	2400,10s	3000,10s	3250,10s	3900,10s	4900,10s

2. For three-phase inverters(X3-Hybrid-G4 as an example), the output power of **EPS** ports is restricted to half (50%) of the nominal output power of **EPS** ports (with battery) total phase, but there are some differences about per-phase power. For more details in the Table2 below.

Table2. EPS output of X3-Hybrid G4 (without battery)

EPS output (without battery)	X3 Hybrid 5.0	X3 Hybrid 6.0	X3 Hybrid 8.0	X3 Hybrid 10.0	X3 Hybrid 12.0	X3 Hybrid 15.0
Nominal output power [W]	2500	3000	4000	5000	6000	7500
Per-phase power [W]	2500	3000	4000	5000	5000	5000
Peak apparent power [VA] (130% overload)	3250,10s	3900,10s	5200,10s	6500,10s	7800,10s	9570,10s

Notes:

- ① If PV input power is sufficient, there will not be any trouble. But if PV input power is not enough, an "EPS Overload" will be reported on the LCD.

Advantages

Some customers install the hybrid inverter first as a string inverter (**Battery Ready**) and add the batteries later, now they can use the back-up without battery.

Configuration

Setting path:

Main → Settings → Advance Setting → EPS Setting → Super-Backup

==EPS setting==(379)
Super-Backup: "(114)
"Enable "(133) / "Disable "(134)"

This function is **Disable** by default.

If users want to supply **EPS** loads only with PV, just **Enable** it.

Compatibility

Hardware version:

Function	Model
Super-Backup	X1-Hybrid G4
	X3-Hybrid G4
	X1-IES

	X3-IES
	X3-ULTRA
	X3-AELIO
	X3-HYB-G4 PRO

Software version (latest version) :

Model	ARM	DSP
X1-Hybrid G4	≥1.44	≥1.48
X3-Hybrid G4	≥1.44	≥1.47
X1-IES	≥009.12	≥013.12
X3-IES	≥012.05	≥016.04
X3-ULTRA	≥018.02	≥017.05
X3-AELIO	≥004.04	≥004.05
X3-HYB-G4 PRO	≥005.11	≥005.12